

Étienne Fodor

Physics of Active Matter

Associate Professor

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Dept of Physics and Materials Science (DPhyMS)

Univ of Luxembourg

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Scientific positions and education

Since 2025	Associate Professor , DPhyMS, Univ of Luxembourg
2020–25	Assistant Professor , DPhyMS + ATTRACT Fellow , Fonds National de la Recherche (FNR)
2017–20	Oppenheimer Fellow , Univ of Cambridge + Junior Research Fellow , St Catharine’s College
2016–17	Postdoctoral Research Associate , DAMTP, Univ of Cambridge (Supervisor: ME Cates)
2013–16	PhD in Physics , Univ Paris Diderot (Supervisors: P Visco, F van Wijland), summa cum laude PhD thesis: “Tracking nonequilibrium in living matter and self-propelled systems”
2012–13	Masters in Physics , École Normale Supérieure (ENS), Paris
2011–12	Agrégation de Physique , Training program for teaching in Physics Bachelors, ENS Cachan
2009–11	Bachelors + Masters in Physics , ENS Lyon

Research and teaching experience

Since 2022	Masters course in Physics , DPhyMS, Univ of Luxembourg 14 weeks/year Course: “Nonequilibrium soft and active matter”
2024	Masters course in Physics , Dept of Physics, Univ of Liège 10 hours, 1 week Course: “Statistical physics of passive and active matter”
2021	Doctoral course in Physics , DPhyMS, Univ of Luxembourg 6 hours, 1 day Course: “Field theory for soft matter”
2015–16	PhD exchange program , YITP, Kyoto Univ (Host: H Hayakawa) 2 months/year
2013–16	Bachelors Physics tutorials , Univ Paris Diderot 14 weeks/year
2013	Masters internship , Univ Paris Diderot (Supervisors: P Visco, F van Wijland) 16 weeks Masters thesis: “Active forces in living cells”
2012–13	Bachelors Physics tutorials , Lycée Fénélon, Paris 30 weeks
2011	Masters intership , Univ of Oxford (Supervisors: AS Wyatt, IA Walmsley) 12 weeks Masters thesis: “Characterization and control of extreme pulses from high harmonic generation”
2010–11	Bachelors Physics tutorials , Lycée la Martinière Monplaisir, Lyon 30 weeks
2010	Bachelors internship , Univ de Genève (Supervisors: L Bonacina, J-P Wolf) 8 weeks Bachelors thesis: “Micro-spectroscopie CARS à l’aide d’une source laser unique”

Funding and selected awards

2027	Outstanding Junior Fellowship , Institute of Advanced Studies, Tel Aviv Univ
2026	Emerging Investigator , Special issue at Soft Matter + Visiting Scholar , ESPCI, Paris
2026–29	CORE grant, OptimalControlTransition , FNR, Luxembourg 800 kEUR Doctoral Training Unit, BECR , FNR (Program lead: A Skupin) Doctoral Training Network, CAFE-BIO , Horizon Europe (Program lead: R Blythe)
2025	Lorentz Center Workshop, Nonequilibrium Systems Under Control , Leiden 30 kEUR
2024–27	CORE grant, ActiveTopo , FNR, Luxembourg 830 kEUR
2021–26	Doctoral Training Unit, ACTIVE , FNR (Program lead: M Esposito)
2020–25	ATTRACT Fellowship , FNR 1.5 MEUR
2017–20	Oppenheimer + Junior Research Fellowship , St Catharine’s College, Univ of Cambridge
2017	PhD prize , Institut Systèmes Complexes, Paris + Best talk , SIAM-IMA Conference, Cambridge
2015	Best talk , Active Liquids, Lorentz Center, Leiden
2013–16	Teaching Assistantship , Univ Paris Diderot + PhD Scholarship , ENS Cachan

Supervision and mentoring

Since 2020	Principal Investigator , Head of group “Physics of active matter”, Univ of Luxembourg
8 Postdocs	Since 2025 A Soriani + F Thewes
	2024–25 F Serafin (after: Postdoc, Univ of Luxembourg)
	2023–25 UA Dattani (after: Deutsche Bank, Berlin)
	2022–26 T Banerjee, CORE Junior Fellow (after: Faculty member, NISER, India)
	2022–25 WD Piñeros, MSCA Fellow (after: Postdoc, CEA Caen, France)
	2021–24 A Manacorda, MSCA Fellow (after: Postdoc, Sapienza Univ, Roma)
	2020–22 LK Davis (after: Flora Philip Research Fellow, Univ of Edinburgh)
7 PhDs	Since 2026 A Jagannathan
	Since 2024 N Setzkorn + IJC Miranda + M Antonioli
	Since 2023 L Casagrande, AFR Fellow (Luxembourg)
	2021–25 BN Radhakrishnan (after: Postdoc, Western Univ, Canada)
	2020–23 Y Zhang (after: United Consulting Group, Frankfurt)
4 Masters	2025–26 A Jagannathan (after: PhD in my group)
	2023–24 N Setzkorn (after: PhD in my group)
	2022–23 L Casagrande (after: PhD in my group) + T Desaleux (after: PhD, France)
2016–20	Student co-supervisor , DAMTP, Univ of Cambridge (supervisor: ME Cates)
2 PhDs	ØL Borthne (2017–20) + T Ekeh (2018–21)
3 Masters	D Martin (2016–17) + T Ekeh (2017–18) + JW Knight (2019–20, best thesis prize)

PhD committees

2026	Univ of Luxembourg , A Kabylda* (supervisor: A Tkatchenko)
	Univ of Barcelona , Y Rouzaire (supervisor: D Levis)
	Tata Institute of Fundamental Research , A Bansal (supervisor: M Rao)
2025	Univ of Luxembourg , SGM Srinivas* (supervisor: M Esposito)
	Univ of Mons , G Palumbo (supervisor: P Damman)
	Univ of Luxembourg , M Puleva* (supervisor: A Tkatchenko)
2024	Univ Grenoble Alpes , L Guislain (supervisor: E Bertin)
	Univ Paris Cité , A Dinelli, (supervisor: J Tailleur)
	Univ of Luxembourg , N Carabba* + L Dupays (supervisor: A del Campo)
2023	Imperial College London , Z Zhang (supervisor: G Pruessner)
2022	Univ of Luxembourg , E Penocchio + D Forastiere (supervisor: M Esposito)
	Univ of Luxembourg , V Vassilev Galindo (supervisor: A Tkatchenko)
2021	Luxembourg Centre for Systems Biomedicine , S Martina (supervisor: A Skupin)
	Univ of Luxembourg , J Ekström* (supervisor: TL Schmidt)

* Head of jury for these PhD defences

Review and outreach activities

Since 2025	Editorial board member , Journal of Statistical Mechanics
Since 2016	Reviewer for journals and agencies ca 30 reviews/year
	Selected journals Commun Phys, J Chem Phys, J Phys A, J Stat Mech, Nat Commun, Nat Phys, New J Phys, Newton, Phys Rev (E, Lett, Res, X), PNAS, Science, Science Adv
	Agencies ANR (France), DFG (Germany), DOE (USA), FRS-FNRS (Belgium), ISF (Israel) (EU)
2026	Guest editor , Physical Review E Special topics: “Controlling stochastic dynamics across scales” Co-editor: TR Gingrich (Northwestern Univ)
2026	Guest editor , New Journal of Physics Special topics: “Statistical mechanics of active matter” Co-editors: M Fruchart (ESPCI), SAM Loos (MPI Goettingen), T Markovich (Tel Aviv Univ)
2024	Outreach lecture , Institut Etudes Scientifiques, Cargèse
2023	Inaugural lecture , Faculty of Science, Technology and Medicine, Univ of Luxembourg
2021–24	High-school seminars and intern hosts + Student fairs , DPhyMS, Univ of Luxembourg

Organized events

- 2026 **Nonequilibrium Statistical Mechanics**, Univ Paris Cité, Paris | Workshop, 1 day
Co-organizer: F van Wijland (Univ Paris Cité)
- 2025 **Nonequilibrium Systems Under Control**, Lorentz Center, Leiden | Workshop, 1 week
Co-organizers: TR Gingrich (Northwestern Univ), SAM Loos (Univ of Cambridge)
- 2024 **Energy, Information and Evolution in Biology**, Cargèse | Summer school, 2 weeks
Co-organizers: A Manacorda, M Esposito (Univ of Luxembourg)
Physics Meets Mathematics, Univ of Luxembourg | Workshop, 1 day
- 2018–20 **Statistical Physics and Soft Matter**, DAMTP, Univ of Cambridge | Weekly seminar
Co-organizers: ME Cates, RL Jack (Univ of Cambridge)
- 2019 **Colloids as a Toolbox for Statistical Mechanics**, Univ of Cambridge | Workshop, 1 day
Co-organizers: ME Cates, RL Jack (Univ of Cambridge)
- 2018 **Nonequilibrium Biophysics, World Congress of Biomechanics**, Dublin | Session, $\frac{1}{2}$ day
Co-organizer: D Mizuno (Kyushu Univ)

Presentation in conferences and seminars

Recording of selected presentations: efodorphysics.github.io/talks.html

Invited presentations in conferences

- 2027 **Mechanical Response and Synchronization in Active Solids**, Cargèse
- 2026 **Nonequilibrium Dynamics, from Active Matter to Evolutionary Dynamics**, KITP
Control and Optimisation of Out-of-equilibrium Processes, Beg Rohu
SigmaPhi2026, Statistical Physics of Complex Systems, Kolymbari
Frontiers in Statistical Physics and Soft Matter, Edwards Symposium, Univ of Cambridge
- 2025 **Statistical Physics of Living Systems**, CECAM, Lausanne
International Discussion Meeting on Relaxations in Complex Systems, Barcelona
Self-Organization Far From Equilibrium, APS March meeting, Anaheim
Machine Learning for Enhanced Sampling of Atomistic Systems, Berkeley
- 2024 **The Many Faces of Active Mechanics**, KITP, Santa Barbara
Nonequilibrium Statistical Physics of Complex Systems, Seoul
- 2023 **Computational Advances in Active Matter**, Lorentz Center, Leiden
Frontiers in Nonequilibrium Physics: Active Matter, Topology and Beyond, Kyoto
Conference on Statistical Mechanics, Sitges
Physics of Dense and Active Disordered Materials, Kyoto
Frontiers in Nonequilibrium Physics, Institute of Mathematical Sciences, Chennai
- 2022 **Statistical Mechanical Theories of Emergence in Biological Systems**, Edinburgh
Numerical Techniques for Nonequilibrium Steady States, CECAM, Mainz
- 2020 **Symmetry, Thermodynamics and Topology in Active Matter**, KITP (online)
- 2018 **Why Measure Entropy Production?**, Princeton Univ
Active Matter Session, Berkeley

Contributed presentations in conferences

- 2026 **DPG Spring Meeting, Active Matter V**, Dresden
- 2025 **Multiscale Modeling of Chemically Active Mixtures**, CECAM, Lausanne
StatPhys29, Out-of-equilibrium Statistical Physics, Florence
From Flocking Birds to Migrating Cells: Recent Advances in Active Matter, Leiden
- 2024 **Dissipative Processes in Molecular Systems**, Padova
Workshop on Stochastic Thermodynamics V (online)
DPG Spring Meeting, Stochastic Thermodynamics, Berlin
- 2023 **StatPhys28, Soft Matter**, Tokyo
Bridge between Non-equilibrium Statistical Physics and Biology, Cambridge
New Perspectives in Active Systems, Dresden
From Soft Matter to Biophysics, Les Houches

2021 **Liquid Matter Conference**, Prague (online)
Workshop on Stochastic Thermodynamics II (online)

2020 **Motile Active Matter Conference**, Bonn (online)

2019 **StatPhys27, Out-of-equilibrium aspects**, Buenos Aires
International Soft Matter Conference, Edinburgh
Statistical Physics of Complex Systems, Nordita, Stockholm

2018 **Nonequilibrium Collective Dynamics**, Technische Univ Berlin
Fundamental Problems in Active Matter, Aspen Center for Physics

2017 **SIAM-IMA Annual Conference**, Univ of Cambridge
Edwards Centre Mini Conference, Univ of Cambridge
Open Statistical Physics, Milton Keynes

2016 **StatPhys26, Biological Physics**, Lyon
Non-Gaussian Workshop, YITP, Kyoto

2015 **Active Liquids**, Lorentz Center, Leiden

2014 **Condensed Matter in Paris**, Univ Paris Descartes
Journées de Physique Statistique, Paris

Invited seminars and colloquia

2027 **Physics Dept Seminar**, Technion
Institute of Advanced Studies Colloquium, Tel Aviv University

2026 **Center for Data Science and Complexity Colloquium**, Univ of Münster
Institut für Theoretische Physik Seminar, Univ of Münster
Gulliver Lab Seminar, ESPCI, Paris
MPI Physics of Complex Systems Seminar, Dresden
Dept of Physics Seminar, Univ of Oxford
Theoretical Biophysics Seminar, Curie Institute, Paris

2025 **Soft Condensed Matter Seminar**, Harvard University
Biophysics Seminar, Massachusetts Institute of Technology
Solitons at Work Seminar (online)

2024 **Dept of Chemistry Seminar**, Univ of California, Berkeley
LPTMC Seminar, Sorbonne Univ, Paris
Dept of Physics Seminar, Univ of Liège
Institute of Physics Seminar, Univ of Leiden
Niels Bohr Institute Seminar, Univ of Copenhagen

2023 **Biological, Soft and Complex Materials and Theory Seminar**, Univ of Bristol
EMBL Theory Seminar, Heidelberg

2022 **Biological Physics and Physical Biology Seminar** (online)
Soft Matter Seminar, DAMTP, Univ of Cambridge (online)
Mathematical Physics Seminar, Imperial College London (online)

2021 **Dept of Physics**, Guangdong Technion (online)
Quantum Science and Technology Seminar, Univ of Luxembourg (online)
Non-equilibrium Statistical Physics Seminar, Georg-August-Univ Göttingen (online)
Centre de Physique Théorique Seminar, Aix-Marseille Univ (online)

2020 **School of Physics and Astronomy Seminar**, Univ of Edinburgh (online)
Dept of Physics Seminar, Univ of Bath

2019 **Quantitative Life Sciences Seminar**, ICTP, Trieste
Dept of Chemistry Seminar, James Franck Institute, Univ of Chicago
Physics of Living Systems Seminar, Massachusetts Institute of Technology
DPhyMS Seminar, Univ of Luxembourg
Institute of Physics Seminar, Univ of Amsterdam

2018 **LiPhy Seminar**, Univ Grenoble Alpes
Charles Coulomb Lab Seminar, Univ de Montpellier

- Gulliver Lab Seminar, ESPCI, Paris
- Graduate Research Seminar, St Catharine's College, Cambridge
- Research Colloquium, California State Univ, Fullerton
- 2017 **Soft Matter Seminar**, DAMTP, Univ of Cambridge
- BioLunch Seminar**, DAMTP, Univ of Cambridge
- 2016 **School of Mathematical Sciences Seminar**, Queen Mary Univ of London
- Soft Matter Seminar**, DAMTP, Univ of Cambridge
- MSC Lab Seminar**, Univ Paris Diderot
- YITP Seminar**, Kyoto
- 2015 **LiPhy Seminar**, Univ Grenoble Alpes
- Physics-Biology Interface Seminar**, Univ Paris Sud
- Soft Matter Seminar**, DAMTP, Univ of Cambridge
- YITP Seminar**, Kyoto
- 2014 **MSC Lab Seminar**, Univ Paris Diderot

Scientific manuscripts

Main publications: Phys Rev Lett [40][35][34][32][7], Phys Rev X [36][25][18][17][9], Reviews [28][24][13]

- [50] **Topology of pulsating active matter: Defect asymmetry controls emergent motility**
L Casagrande, A Manacorda, and ÉF, arXiv:2605.25996
- [49] **Collective deformation of anisotropic particles with internal pulsation**
L Casagrande, A Manacorda, and ÉF, arXiv:2605.25744
- [48] **Contraction waves in pulsating active liquids: From pacemaker to aster dynamics**
T Banerjee, T Desaleux, J Ranft, and ÉF, arXiv:2509.19024
- [47] **Control of active field theories at minimal dissipation**
A Soriani, E Tjhung, ÉF, and T Markovich, arXiv:2504.19285
- [46] **Hydrodynamics of pulsating active liquids**
T Banerjee, T Desaleux, J Ranft, and ÉF, arXiv:2407.19955
- [45] **Irreversibility in scalar active turbulence: The role of topological defects**
BN Radhakrishnan,* F Serafin,* TL Schmidt, and ÉF, New J Phys **28**, 034601 (2026)
- [44] **Entropy production rate in thermodynamically consistent flocks**
T Agranov, RL Jack, ME Cates, and ÉF, New J Phys **27**, 104602 (2025)
- [43] **Quantifying dissipation in flocking dynamics: When tracking internal states matters**
K Proesmans, G Falasco, A Tanaji Mohite, M Esposito, and ÉF, Phys Rev E **112**, 024103 (2025)
- [42] **Diffusive oscillators capture the pulsating states of deformable particles**
A Manacorda and ÉF, Phys Rev E **111**, L053401 (2025)
- [41] **Species interconversion of deformable particles yields transient phase separation**
Y Zhang, A Manacorda, and ÉF, New J Phys **27**, 043023 (2025)
- [40] **Biased ensembles of pulsating active matter**
WD Piñeros and ÉF, Phys Rev Lett **134**, 038301 (2025) | Editors' suggestion
- [39] **Nonequilibrium thermodynamics of non-ideal reaction-diffusion systems: Implications for active self-organization**
F Avanzini, T Aslyamov, ÉF, and M Esposito, J Chem Phys **161**, 174108 (2024)
- [38] **Controlling active matter: The need for thermodynamic consistency**
ÉF, Europhys News **55**, 20 (2024)
- [37] **Thermodynamically consistent flocking: From discontinuous to continuous transitions**
T Agranov, RL Jack, ME Cates, and ÉF, New J Phys **26**, 063006 (2024)
- [36] **Active matter under control: Insights from response theory**
LK Davis, K Proesmans, and ÉF, Phys Rev X **14**, 011012 (2024) | Highlight in Physics 17, 20 (2024)

- [35] **Pulsating active matter**
Y Zhang and ÉF, Phys Rev Lett **131**, 238302 (2023)
- [34] **Non-ideal reaction-diffusion systems: Multiple routes to instability**
T Aslyamov, F Avanzini, ÉF, and M Esposito, Phys Rev Lett **131**, 138301 (2023)
- [33] **Towards a liquid-state theory for active matter**
YI Li, R Garcia-Millan, ME Cates, and ÉF, EPL **142**, 57004 (2023)
- [32] **Thermodynamic control of activity patterns in cytoskeletal networks**
A Lamtyugina, Y Qiu, ÉF, AR Dinner, and S Vaikuntanathan, Phys Rev Lett **129**, 128002 (2022)
- [31] **From predicting to learning dissipation from pair correlations of active liquids**
G Rassolov, L Tociu, ÉF, and S Vaikuntanathan, J Chem Phys **157**, 054901 (2022)
- [30] **Mean-field theory for the structure of strongly interacting active liquids**
L Tociu, G Rassolov, ÉF, and S Vaikuntanathan, J Chem Phys **157**, 014902 (2022)
- [29] **Power fluctuations in sheared amorphous materials: A minimal model**
T Ekeh, ÉF, SM Fielding, and ME Cates, Phys Rev E **105**, L052601 (2022)
- [28] **Irreversibility and biased ensembles in active matter: Insights from stochastic thermodynamics**
ÉF, RL Jack, and ME Cates, Annu Rev Condens Matter Phys **13**, 215 (2022)
- [27] **Stochastic hydrodynamics of complex fluids: Discretisation and entropy production**
ME Cates, ÉF, C Nardini, T Markovich, and E Tjhung, Entropy **24**, 254 (2022) | Editor's choice
- [26] **Optimal power and efficiency of odd engines**
ÉF and A Souslov, Phys Rev E **104**, L062602 (2021)
- [25] **Thermodynamics of active field theories: Energetic cost of coupling to reservoirs**
T Markovich, ÉF, E Tjhung, and ME Cates, Phys Rev X **11**, 021057 (2021)
- [24] **Active engines: Thermodynamics moves forward**
ÉF and ME Cates, EPL **134**, 10003 (2021)
- [23] **Statistical mechanics of active Ornstein-Uhlenbeck particles**
D Martin, J O'Byrne, ME Cates, ÉF, C Nardini, J Tailleur, and F van Wijland, Phys Rev E **103**, 032607 (2021)
- [22] **Collective motion in large deviations of active particles**
Y-E Keta, ÉF, F van Wijland, ME Cates, and RL Jack, Phys Rev E **103**, 022603 (2021)
- [21] **Time-reversal symmetry violations and entropy production in field theories of polar active matter**
ØL Borthne, ÉF, and ME Cates, New J Phys **22**, 123012 (2020)
- [20] **Thermodynamic cycles with active matter**
T Ekeh, ME Cates, and ÉF, Phys Rev E **102**, 010101(R) (2020)
- [19] **Dissipation controls transport and phase transitions in active fluids: Mobility, diffusion and biased ensembles**
ÉF, T Nemoto, and S Vaikuntanathan, New J Phys **22**, 013052 (2020)
- [18] **Autonomous engines driven by active matter: Energetics and design principles**
P Pietzonka, ÉF, C Lohrmann, ME Cates, and U Seifert, Phys Rev X **9**, 041032 (2019)
- [17] **How dissipation constrains fluctuations in nonequilibrium liquids: Diffusion, structure and biased interactions**
L Tociu, ÉF, T Nemoto, and S Vaikuntanathan, Phys Rev X **9**, 041026 (2019)
- [16] **Driven probe under harmonic confinement in a colloidal bath**
V Démery and ÉF, J Stat Mech **2019**, 033202 (2019)
- [15] **Optimizing active work: Dynamical phase transitions, collective motion and jamming**
T Nemoto, ÉF, ME Cates, RL Jack, and J Tailleur, Phys Rev E **99**, 022605 (2019)

- [14] **Non-Gaussian noise without memory in active matter**
ÉF, H Hayakawa, J Tailleur, and F van Wijland, Phys Rev E **98**, 062610 (2018)
- [13] **The statistical physics of active matter: From self-catalytic colloids to living cells**
ÉF and MC Marchetti, Physica A **504**, 106 (2018)
- [12] **Extracting maximum power from active colloidal heat engines**
D Martin, C Nardini, ME Cates, and ÉF, EPL **121**, 60005 (2018) | Editor's choice
- [11] **Active mechanics reveal molecular-scale force kinetics in living oocytes**
WW Ahmed,* ÉF,* M Almonacid,* M Bussonnier, NS Gov, M-H Verlhac, P Visco, F van Wijland, and T Betz, Biophys J **114**, 1667 (2018)
- [10] **Spatial fluctuations at vertices of epithelial layers: Quantification of regulation by Rho pathway**
ÉF,* V Mehandia,* J Comelles, R Thiagarajan, NS Gov, P Visco, F van Wijland, D Riveline
Biophys J **114**, 939 (2018)
- [9] **Entropy production in field theories without time-reversal symmetry: Quantifying the non-equilibrium character of active matter**
C Nardini, ÉF, E Tjhung, F van Wijland, J Tailleur, and ME Cates, Phys Rev X **7**, 021007 (2017)
- [8] **Nonequilibrium dissipation in living oocytes**
ÉF,* WW Ahmed,* M Almonacid,* M Bussonnier, NS Gov, M-H Verlhac, T Betz, P Visco, and F van Wijland, EPL **116**, 30008 (2016)
- [7] **How far from equilibrium is active matter?**
ÉF, C Nardini, ME Cates, J Tailleur, P Visco, and F van Wijland, Phys Rev Lett **117**, 038103 (2016)
Editor's suggestion | Highlight in Physics 9, s76 (2016)
- [6] **Active cage model of glassy dynamics**
ÉF, H Hayakawa, P Visco, and F van Wijland, Phys Rev E **94**, 012610 (2016)
- [5] **Modeling the dynamics of a tracer particle in an elastic active gel**
E Ben Isaac, ÉF, P Visco, F van Wijland, and NS Gov, Phys Rev E **92**, 012716 (2015)
- [4] **Active cell mechanics: Measurement and theory,**
WW Ahmed, ÉF, and T Betz, Biochimica et Biophysica Acta - Mol Cell Res **1853**, 3083 (2015)
- [3] **Activity-driven fluctuations in living cells**
ÉF,* M Guo,* NS Gov, P Visco, DA Weitz, and F van Wijland, EPL **110**, 48005 (2015)
Editor's choice | Highlight in Europhysics News 46/5 (2015)
- [2] **Generalized Langevin equation with hydrodynamic backflow: Equilibrium properties**
ÉF, DS Grebenkov, P Visco, and F van Wijland, Physica A **422**, 107 (2015)
- [1] **Energetics of active fluctuations in living cells**
ÉF, K Kanazawa, H Hayakawa, P Visco, and F van Wijland, Phys Rev E **90**, 042724 (2014)

* Equal contribution of these authors to this work